FIT2093 Assignment 1 (Task 1) – Semester 1, 2024
Submission Guidelines & Tasks

Guidelines Details

- **Deadline:** Assignment 1 **Task 1** is due on **28 March 2024** at **11:55pm** (Melbourne time). This Task is an individual (not group) work and it must be submitted by each student **individually**.
- **Submission Platform:** Electronic submission via the Assignment 1 Task 1 submission link on the Moodle Assessments page (an Ed announcement will be sent once the submission link is added).
- **Required Files:** Required data and numerical parameter files for the assignment will be available for download via the Assignment 1 Task 1 Specification Download link on the Moodle Assessments page.
- **Submission File Format:** Two PDF documents: one for SubTask 1.1 and one for SubTask 1.2; as well as accompanying video clips (optional).
- **Submission Page Limit:** Each submitted PDF document must be **at most 7 pages (including screenshots)**, excluding the cover page and references.
- **Plagiarism:** It is an academic requirement that your submitted work be original. Zero marks will be awarded for the whole submission if there is any evidence of copying, collaboration, pasting from web sites, or copying from textbooks.
- **Use of Generative AI tools:** ChatGPT or other AI tools may be used for study purposes, to learn about your topic, and to develop your assignment. However, similar to citation requirements for references, you must include a clear declaration of all generative AI tools used (e.g. ChatGPT, DALL-E, Grammarly, voice-to-text), how and where you have used them.

Notes

1. You can use the `SageMath` tool to perform any calculation necessary for this assignment. The `sagemath` web interface is available at: `https://sagecell.sagemath.org/`. Refer to the Applied Session exercises.

2. **For each question**, you need to answer **both**
   i. the computation **result** question
   ii. the **explanation** questions about your working process such as the source code or the commands you are using to solve the tasks.

3. Note that if numbers in this assignment are specified in hexadecimal format, your written answer and many software packages expect hexadecimal numbers to be input with a ‘0x’ prefix (e.g. ’0xa0b1c2d3’) for indicating the hexadecimal format. For example, this prefix allows SageMath to interpret the value in hex.

Marks

- This Assignment 1 Task 1 is weighted **10%** of the total unit marks.
- This Task 1 is marked out of **100** nominal marks. Each mark in Task 1 is thus worth 0.1% of the total unit marks.

Task 1 (100 marks)	Marks
SubTask 1.1 (60 marks)	• **Manual** computation (15 marks) • **SageMath** commands & answer (20 marks) • **Explanation** of each operation in both encryption and decryption (15 marks) • **Presentation** (10 marks)
SubTask 1.2 (40 marks)	• **Evaluation** of different rounds for the given inputs (15 marks) • **Explanation** the importance and the relationship of these operations (15 marks) • **Visual** aids and presentation (10 marks)

Overview

A software development company called **SecureApps** is developing ***PeerHelper***, a secure messaging application used by several different communities for in-community peer-to-peer mental health support. The application allows direct messages, where both parties can see the message history from any of their authenticated devices for up to an hour, after which they disappear. They can also see a number of group threads and chats which are persistent.

Due to the potential sensitivity of discussions in the platform, new users must be verified by an existing user from the community they wish to participate in, and confidentiality and authentication security are a priority.

In this assignment, you will investigate how the cryptographic techniques you studied in this unit can be applied to the development of the PeerHelper App.

Task 1: Investigation of Block Cipher for *PeerHelper* Message Confidentiality (10%)

To protect message confidentiality from group outsiders, the ***PeerHelper*** software developers intend to use a block cipher in a specific mode of operation. They are considering the use of a block cipher called **Canary,** whose block diagram is described in **Fig. 1**. Your task is to help the ***PeerHelper*** developers to **understand** how Canary works and **examine** its behaviour.

SubTask 1.1

In order to make a case for this cipher, you first need to explain this cipher to the ***PeerHelper*** software developers (Kiran Jain and Misao Nakano).

1.1 A) Run through the three operations of one round **by hand** (*showing your working in computing the output of each of the round operations*), and use **SageMath** commands to encrypt the message provided in your Moodle Assessments page *Assignment 1 Task 1 Individual Specification* link, with the provided key.

Hint: Use suitable variants of the SageMath commands you have learnt in the Week 3 Applied Session.

1.1 B) **Draw** the block diagram for decrypting one round of the Canary block cipher, and use it to **decrypt** the ciphertext from 1.1A) using **SageMath** commands, showing the **intermediate results** after each operation until you reach the input message.

1.1 C) Using all your work from the above steps and the block diagram for encryption in Fig 1, **write** up an **email** to Kiran and Misao explaining **how** encryption and decryption works in the Canary cipher, justifying the **purpose of each step** for the overall security. *You may also include a short video clip with your facial inset to accompany your descriptions of the cipher for a more personalized quality explanation of what you understand about the cipher*. Be sure to include your own block diagram for the decryption algorithm to support your explanation, and provide **screenshots** of your SageMath commands and **intermediate results** after each of the three operations to illustrate your explanation with an example with the inputs provided to you in 1.1A).

Note: you have spoken with Kiran and Misao before but don't know them well, so your email should be formatted in a more formal way and be easy to read in terms of structure.

Submission: Submit your SubTask 1.1 email (to Kiran and Misao) as a PDF file at the Moodle Assessment Page Assignment 1 Task 1 submission link. *You may also submit your accompanying video clip (maximum 5 minutes long) as a separate file submission on Moodle.*

The Canary block cipher encryption algorithm Enc has a 16-bit block state and consists of N rounds, where N is a parameter of the block cipher, and each round is composed of three operations: **KeyXOR**, **ShiftLeft** and **Sub**, as illustrated in the following **Fig. 1**. The round operations are specified in **Fig. 2** below.

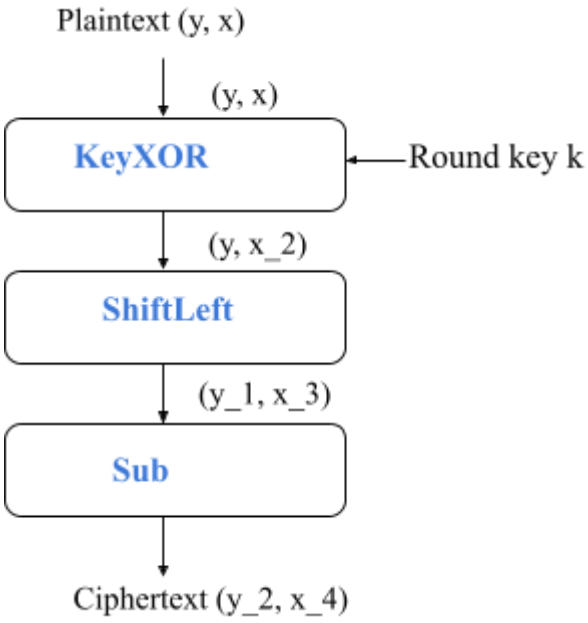


Fig. 1. Illustration of one round of the **Canary block cipher** operations.

Input: The 16-bit plaintext block state (y, x) and the round key k , where:

a. x is the **rightmost** 8-bit of the block state and
b. y be the **leftmost** 8-bit of the block state, i.e. (y, x).
c. k is an 8-bit round key.

1. **KeyXOR**: Compute x_2 = x ⊕ k (where ⊕ denotes the bitwise exclusive or). The new block state after this operation is (y, x_2). For example,
   if x = D3H = 11010011B and k = 74H = 01110100B in hex/binary,
   then x_2 = x ⊕ k = D3H ⊕ 74H = 11010011B ⊕ 01110100B = 10100111B = A7H.

2. **ShiftLeft:** Perform a **2-bit** circular left-shift **of the whole 16-bit block state** (y, x_2). The new block state after this operation is (y_1, x_3). For example, if y = 74H = 01110100B and x_2 = A7H = 10100111B in hex/binary, then (y_1, x_3) = (01110100 || 10100111) ≪ 2 = 1101**0010** 1001**1101** = (D2H, 9DH), where || denotes concatenation. See Fig. 3 for an example of a 1-bit circular left shift.

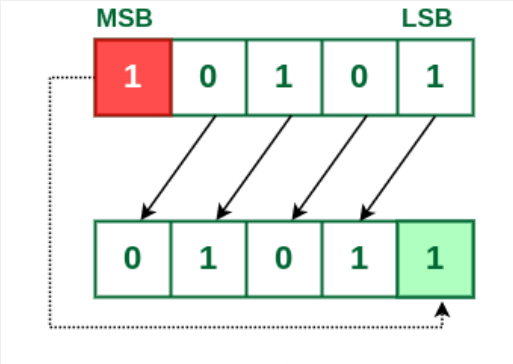


Fig. 3. Example 1-bit left circular shift

3. **Sub:** Perform the 4-bit **substitution table operation on every 4-bit word of the state,** where the substitution table (S-box) is given in **Fig. 4.** The new block state after this operation is (y_2, x_4). For example, if (y_1, x_3) = (D2, 9D) in hex, then (y_2, x_4) = (E0, 6E) in hex.

Output: The 16-bit ciphertext block state (y_2, x_4), where:

a. The **rightmost** 8-bit of the ciphertext block is x_4
b. The **leftmost** 8-bit of the ciphertext block is y_2.

Fig. 2. Specification of the **Canary** block cipher encryption round.

Input (hex)	Output (hex)
0	7
1	F
2	0
3	9
4	3
5	1
6	D
7	B
8	4
9	6
A	2
B	8
C	C
D	E
E	5
F	A

Fig. 4. 4-bit Substitution table (S-box) for Canary block cipher.

SubTask 1.2

Now that you have explained the operation of the Canary block cipher to Kira and Misao, the project engineering manager Reyes Crocifissa has asked you to investigate some **security properties** of Canary and **how many rounds** of the Canary cipher would be necessary for security. Kira and Misao sent you their program implementing the Canary block cipher round, that you can use for your security evaluation. You can find the program and required parameters available for download in *Assignment 1 Task 1 Individual Specification* link on the Moodle Assessment page, along with instructions on how to run it in your VM.

1.2 A). Using Kira and Misao's program, investigate the pseudorandomness security of the Canary blockcipher experimentally.

Step 1: Examine the **outputs** of N = 1 round of the Canary cipher on the following inputs m_0, m_1, m_2, m_3, m_4 with key K = (k_1, k_2, k_3, k_4, k_5). Here, k_i denotes the round key for the i'th round for i=1,2,3,4,5.

Step 2: Repeat the experiment with N = 2, 3, 4, 5 rounds of the cipher respectively, applied to the same inputs.

1.2 B). Write a brief report (*please put your name, student ID and casual photo on the cover page*) on your security evaluation of Canary for Reyes, summarising your **experimental results** above. *You may also include a short video clip with your facial inset to accompany your descriptions of your results, for a more personalized quality explanation of analysis and what you observed about the cipher's properties.*

Explain in your report:

- how you evaluated the **pseudorandom function security** of Canary based on:
  - its behaviour with **different** values of the **round** parameter N. Make sure to discuss:
  - your **experimental** results, including the appropriate **screenshots**
  - your **reasoning** and conclusions on the pseudorandom function security
- your explanation of these experimental results, based on:
  - your knowledge of the **block diagram** and internal **operations** of Canary
  - your recommendation for the necessary **minimum** number of rounds N required for security.

***Note**: the length of the key is kept short in Canary for ease of hand computations, you should focus on other security properties besides exhaustive key search).*

Example of input and output for the Canary executable file:

```
fit2093@fit2093-vm:~$ ./Canary_86 -k 0x88,0x46,0x09,0x44,0x77 -s 1218446 -m 0xaa
aa -n 5
Cipher of m_0 (0xaaaa) after 5 round(s): 0x[redacted]
Cipher of m_1 (0xaaa8) after 5 round(s): 0x[redacted]
Cipher of m_2 (0xaaab) after 5 round(s): 0x[redacted]
Cipher of m_3 (0x2aaa) after 5 round(s): 0x[redacted]
Cipher of m_4 (0xeaaa) after 5 round(s): 0x[redacted]
```

Instructions for running the downloaded Canary executable file in VM:

1. In your VM, open the Firefox web browser and log in to your FIT1093/2093 Monash Moodle page
2. Download executable file from **Moodle Asg 1 Task Specification Download** link (on Moodle `Assessments' page)
   a. `Canary_86.zip` for x86 machine
   b. `Canary_M` for M1/M2 machine
3. Save the file into /home/fit2093
   a. x86: run command "`unzip Canary_86.zip`" with password fit2093 to get `Canary_86`
   b. M1/M2: no unzip is required
4. Run the executable in the folder (`Canary_86` for x86 VM and `Canary_M` for Mac M1/M2 VM) Parameters for the command Canary

   ```
   -k <given 5 round keys>
   -s <your student ID>
   -m <given message>
   -n <number of rounds>
   ```

   (note: if you get an error "Permission denied" when you try to run the executable, you may need to first allow execution permission for the file, using the command "`chmod +x Canary_M`" or "`chmod +x Canary_86`".)

Submission: Submit your SubTask 1.2 report as a PDF file at the Moodle Assessment Page Task 1 submission link. *You may also submit your accompanying video clip (maximum 5 minutes long) as a separate file submission on Moodle.*